

Applied NLP

DATA 641

Lecture 2 — NLP Pipeline; Text Extraction and Cleanup

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Text Extraction and Cleanup

Process of extracting raw text from the input data by removing all other non-textual information.

- It is an important step that has implications for all other aspects in the NLP pipeline!
- It can be one of the most time-consuming parts in an NLP project. Also, it is application dependent!

[illegible]

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<p>code=" text-align: justify;">
The book will be around 350 pages. It will be accompanied by a code
repository containing several Jupyter notebooks for all the chapters to
give a walk-through and explain the code in detail. The code base is in
Python and various machine learning and natural language processing
libraries. The book assumes that the readers have a good grasp of
programming but no theoretical and practical knowledge of NLP.
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<p><b>
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<a
href="https://www.oreilly.com/library/view/practical-natural-language/9
81492054047/">

</a>

<h1 style="color: #e74c3c;">Commonly Asked Questions</h1>
<ul>
<li>Can I contribute to the book?
<p>The book is accompanied by open source Jupyter notebooks and demo
applications. If you are a great ML or front-end engineer looking to
build something meaningful you can apply by filling <a
href="https://goo.gl/forms/dofNcW251LX26ajk1">this form</a>. Also refer
to the next question.
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HTML Parsing and Cleanup: Extract parts of text from HTML pages

How can we extract text from an HTML page? If we observe the HTML markup of a typical Stack Overflow question page, we notice that questions and answers have special tags associated with them. We can utilize this information while extracting text from the HTML page.

- While it may seem like writing our own HTML parser is the way to go, for most cases we encounter, it is more feasible to utilize existing libraries such as BeautifulSoup and Scrapy, which provide a range of utilities to parse web pages.

Unicode Normalization: To parse non-textual symbols such as emojis and graphical characters we use Unicode normalization

 U+2191	 U+1F647	 U+2010	 U+FF64	 U+0398	 U+1F49A	 U+263B	 U+056E	 U+0C2C	 U+0CA0
 U+03B3	 U+12CE	 U+1F49C	 U+03BC	 U+1F680	 U+266A	 U+FE36	 U+30FB	 U+10E6	 U+2036
 U+263C	 U+0964	 U+26AC	 U+0939	 U+4E14	 U+09F0	 U+0F4F	 U+2030	 U+30C7	 U+21A9
 U+2018	 U+2033	 U+026A	 U+0DA2	 U+1F639	 U+0394	 U+00F9	 U+27AB	 U+0177	 U+1F9D8

Spelling Correction: This process is far from perfect! You can build your own spell checker using a huge dictionary of words from a specific language

System-Specific Error Correction:

- Extract plain text from PDF documents. While there are several libraries, such as PyPDF and PDFMiner, to extract text from PDF documents, they are far from perfect, and it's not uncommon to encounter PDF documents that can't be processed by such libraries.
- Extract plain text from scanned documents